Challenge Track:
Divide and Translate: Parameter Isolation with Encoder Freezing for Low-Resource Indic NMT

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The MMLoSo 2025 Challenge: Translating Low-Resource Indic Languages

The goal is to translate between:

- Hindi ⇔ Bhili
- Hindi ⇔ Gondi
- Hindi ⇔ Mundari
- English ⇔ Santali

Each direction contains only ~20K sentence pairs, many of which are noisy and inconsistent.

This makes the task challenging due to:

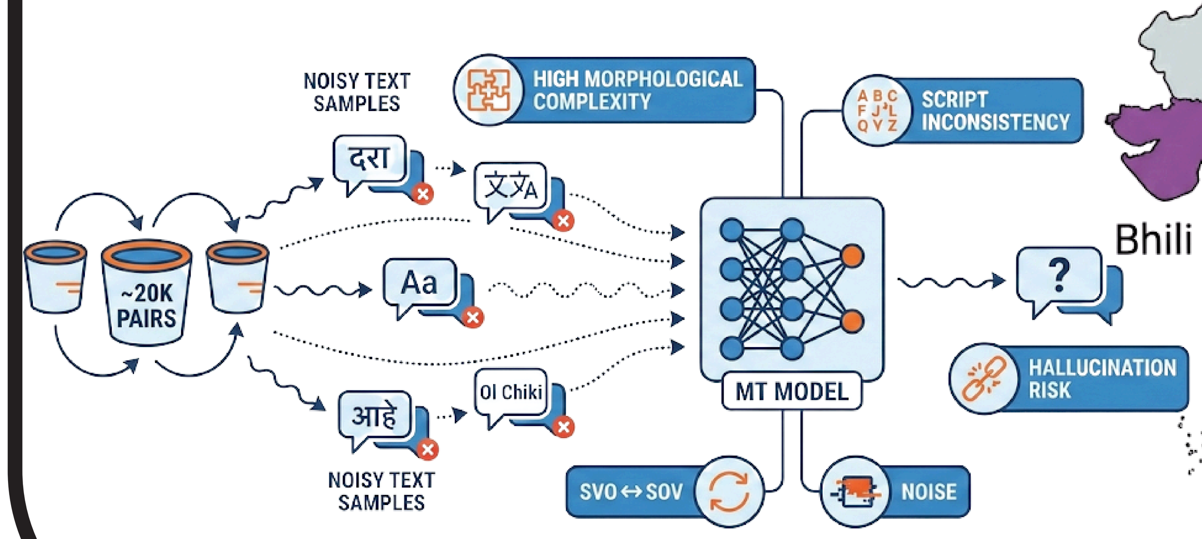
- Severe data scarcity
- High morphological complexity
- Syntactic divergence (SVO vs SOV)
- Script inconsistency (Devanagari, Ol Chiki, Latin)

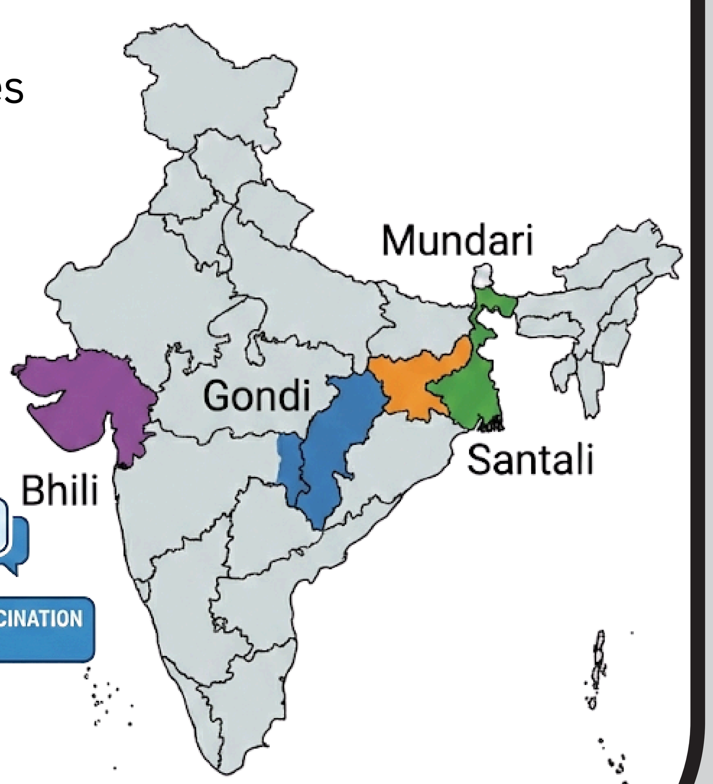
Problem with Unified Multilingual Training

Unified multilingual NMT assumes one shared parameter space can serve all languages. In ultra low resource Indic settings, this assumption breaks.

- Conflicting syntax across language pairs
- Severe data imbalance
- Diluted representation capacity
- Negative transfer from high resource languages

This leads to poor convergence and unstable decoding.





LESSONS FROM FAILED APPROACHES

Strategy	Constraint	Primary Failure Mode
Unified Full FT	Optimization	Gradient Conflict (SVO/SOV)
IndicTrans2 (SOTA)	Domain	Hallucination & Low Recall
LoRA Adapters	Structural	Weak Morphological Modeling

QUALITATIVE TRANSLATION RESULTS

```
Translation English → Santali:
[illegible]

Translation Santali → English:
A terminal will be used to ferry the Shenzhen region.
```

OFFICIAL LEADERBOARD PERFORMANCE

Evaluation Results

Score Type	Leaderboard Result
Public Score	171.4
Private Score	161.1

Less than 6% drop between validation and blind test

This indicates:

- Strong robustness to unseen domains
- Minimal leaderboard overfitting
- Stable performance across data shifts
- Encoder freezing improves generalization consistency

Metric definition:
Weighted score = 0.6 × BLEU + 0.4 × chrF
Per-direction scores were not released by organizers.

DIVIDE AND TRANSLATE FRAMEWORK

Freeze what is universal, specialize what is linguistic.

We preserve multilingual representations using a frozen encoder, and isolate syntactic learning using direction specific decoders.

Data Preprocessing

- Unicode normalization (NFKC)
- Punctuation normalization
- Stratified train validation split (95 / 5)
- Shared SentencePiece tokenizer (V = 256k)

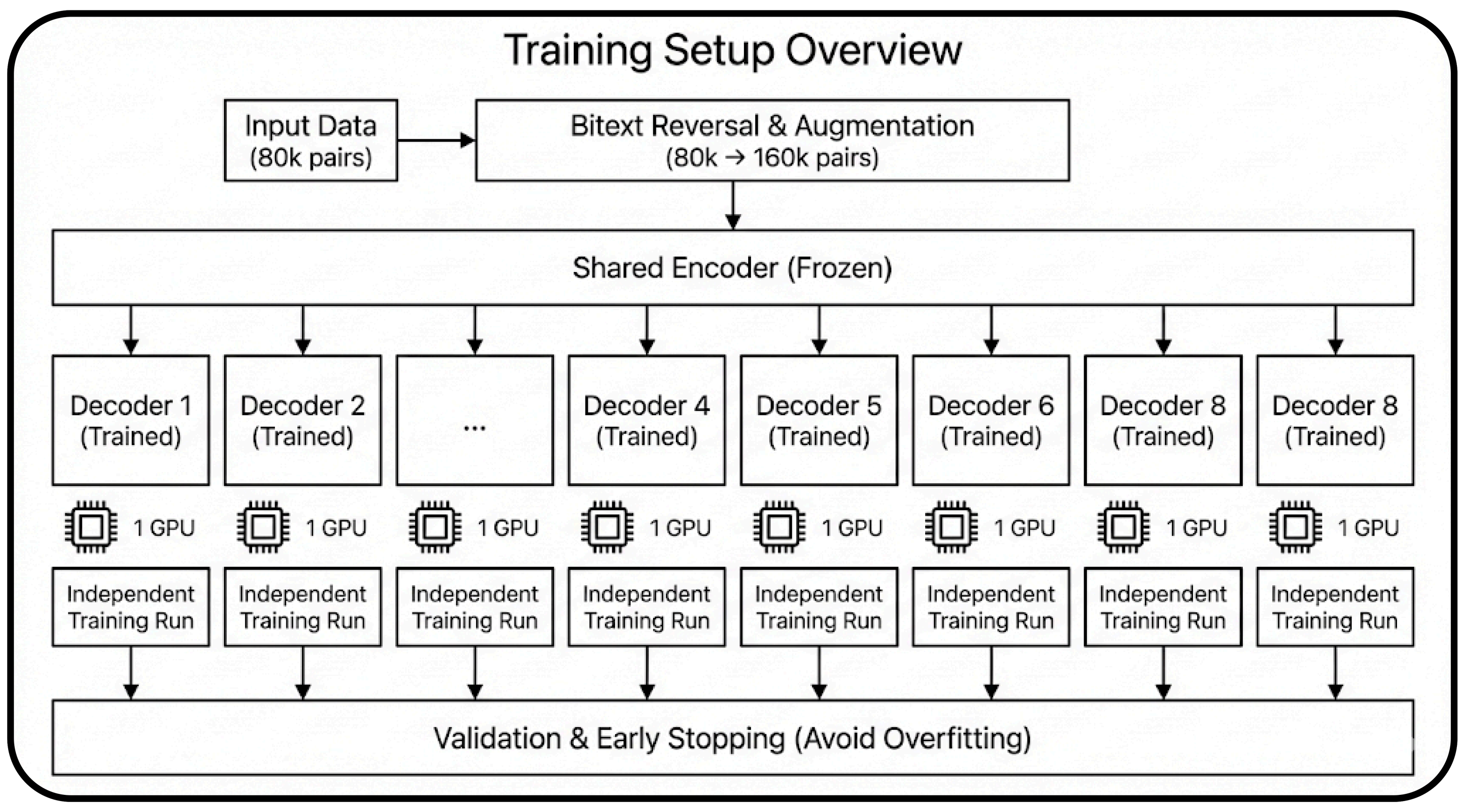
Ensures clean data without aggressive filtering.

Data Augmentation

Bitext Reversal
For each (x, y) pair, we generate (y, x)

- Doubles training data to 160k
- Enables bi directional training
- Improves encoder robustness
- Creates directional symmetry

No synthetic content introduced.



Frozen Encoder

- Based on NLLB 600M pretrained multilingual encoder
- Parameters fully frozen during training
- Preserves cross lingual alignment
- Prevents catastrophic forgetting

Encoder acts as a stable semantic backbone.

Direction Specific Decoders

- One decoder per translation direction
- Eight total experts
- Fully fine tuned
- No parameter sharing across decoders

Each decoder becomes a specialist for one grammar.

Training Setup

Optimization	AdamW, cosine schedule, warmup
Training Duration	3–4 epochs per direction
Stability Choices	Gradient checkpointing
Decoding	Beam search (size = 5)
Hardware	Single GPU (Tesla P100)

→ *Strong results with minimal compute resources*

Key Insights

- Frozen Encoder = Stability
  - No drift.
  - No forgetting.
  - No bias.
- Decoders Become Experts
  - One direction.
  - One expert.
  - Better control.
- Specialization Beats Universality
  - Low data favors isolation.
- Clean Optimization
- Generalization Over Tuning

CONCLUSION

Specialized Experts Outperform Unified Models:

- Divide and Translate shows that expert models excel compared to unified multilingual fine-tuning in extreme low-resource settings with divergent grammars.

Parameter Isolation Reduces Interference:

- Isolating parameters mitigates negative transfer across conflicting source–target structures.

Encoder Freezing Preserves Alignment:

- Freezing the encoder provides strong regularization, maintaining cross-lingual alignment while allowing morphological adaptation.

Balances Fidelity and Adaptability:

- The framework achieves high-quality translation while remaining flexible across languages.

Future Potential:

- Expert models can be distilled into a compact, efficient system without losing the benefits of specialization.

Community Impact:

- This work contributes toward preserving linguistic diversity by enabling digital access for underrepresented communities through reliable translation systems.

"Attention is all you need... but freezing and isolation never hurt."